

Question Paper for 2<sup>nd</sup> Periodical Examination (May 2026) (Session Jan-May 2026)  
 Time: One and a half hours  
 Note: Attempt all questions  
 Max Marks:30

| CO  | COURSE OUTCOMES                                                                                    |
|-----|----------------------------------------------------------------------------------------------------|
| CO1 | Understand fundamentals concepts, models of deep learning.                                         |
| CO2 | Apply deep learning algorithms to learn a model.                                                   |
| CO3 | Implement and analyze various tools to learn efficiently like variants of gradient descent.        |
| CO4 | Apply autoencoders in various applications                                                         |
| CO5 | Apply CNN, RNN and other models to learn from various types of datasets like image, text data set. |

| QNo. | Questions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Marks | CO | KL |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |    |   |   |    |   |   |   |   |   |   |    |    |    |    |   |    |
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| 1.   | Do the following:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |       |    |    |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |    |   |   |    |   |   |   |   |   |   |    |    |    |    |   |    |
| 1.a  | What are skip connections and how are they useful?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 1     | 3  | L1 |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |    |   |   |    |   |   |   |   |   |   |    |    |    |    |   |    |
| 1.b  | Why is regularisation required?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 1     | 3  | L1 |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |    |   |   |    |   |   |   |   |   |   |    |    |    |    |   |    |
| 1.c  | Types of regularisation techniques?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 1     | 3  | L2 |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |    |   |   |    |   |   |   |   |   |   |    |    |    |    |   |    |
| 1.d  | Types of Normalization Techniques?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 1     | 3  | L1 |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |    |   |   |    |   |   |   |   |   |   |    |    |    |    |   |    |
| 1.e  | Explain "Context Vector" in RNN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 1     | 3  | L1 |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |    |   |   |    |   |   |   |   |   |   |    |    |    |    |   |    |
| 2.   | Decompose the following matrix A(2X3) using SVD<br><table border="1" style="margin: 5px;"> <tr><td>1</td><td>1</td><td>0</td></tr> <tr><td>0</td><td>1</td><td>1</td></tr> </table>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 1     | 1  | 0  | 0 | 1 | 1 | 5 | 3 | L2 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |    |   |   |    |   |   |   |   |   |   |    |    |    |    |   |    |
| 1    | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 0     |    |    |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |    |   |   |    |   |   |   |   |   |   |    |    |    |    |   |    |
| 0    | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 1     |    |    |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |    |   |   |    |   |   |   |   |   |   |    |    |    |    |   |    |
| 3.   | What are auto encoders? Explain how PCA is related to this architecture.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 5     | 4  | L3 |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |    |   |   |    |   |   |   |   |   |   |    |    |    |    |   |    |
| 4.   | Explain AlexNet architecture in detail. Compare its shortcomings with other superior architectures.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 5     | 5  | L3 |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |    |   |   |    |   |   |   |   |   |   |    |    |    |    |   |    |
| 5.   | Consider a CNN with 6*6*1 gray scale image and a kernel A and B of size 3*3 is given below:<br><div style="display: flex; justify-content: space-around;"> <div style="border: 1px solid black; padding: 5px;"> <p>Input Feature Map(Gray Scale Image):</p> <table border="1" style="width: 100%;"> <tr><td>1</td><td>2</td><td>3</td><td>0</td><td>1</td><td>2</td></tr> <tr><td>0</td><td>1</td><td>2</td><td>3</td><td>1</td><td>0</td></tr> <tr><td>3</td><td>1</td><td>0</td><td>2</td><td>2</td><td>1</td></tr> <tr><td>2</td><td>2</td><td>1</td><td>0</td><td>1</td><td>3</td></tr> <tr><td>1</td><td>0</td><td>2</td><td>1</td><td>3</td><td>2</td></tr> <tr><td>0</td><td>1</td><td>3</td><td>2</td><td>1</td><td>0</td></tr> </table> </div> <div style="border: 1px solid black; padding: 5px;"> <p>Kernel(A):</p> <table border="1" style="width: 100%;"> <tr><td>1</td><td>0</td><td>-1</td></tr> <tr><td>1</td><td>0</td><td>-1</td></tr> <tr><td>1</td><td>0</td><td>-1</td></tr> </table> <p>Kernel(B):</p> <table border="1" style="width: 100%;"> <tr><td>1</td><td>1</td><td>1</td></tr> <tr><td>0</td><td>0</td><td>0</td></tr> <tr><td>-1</td><td>-1</td><td>-1</td></tr> </table> </div> </div> | 1     | 2  | 3  | 0 | 1 | 2 | 0 | 1 | 2  | 3 | 1 | 0 | 3 | 1 | 0 | 2 | 2 | 1 | 2 | 2 | 1 | 0 | 1 | 3 | 1 | 0 | 2 | 1 | 3 | 2 | 0 | 1 | 3 | 2 | 1 | 0 | 1 | 0 | -1 | 1 | 0 | -1 | 1 | 0 | -1 | 1 | 1 | 1 | 0 | 0 | 0 | -1 | -1 | -1 | 10 | 5 | L4 |
| 1    | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 3     | 0  | 1  | 2 |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |    |   |   |    |   |   |   |   |   |   |    |    |    |    |   |    |
| 0    | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2     | 3  | 1  | 0 |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |    |   |   |    |   |   |   |   |   |   |    |    |    |    |   |    |
| 3    | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 0     | 2  | 2  | 1 |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |    |   |   |    |   |   |   |   |   |   |    |    |    |    |   |    |
| 2    | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 1     | 0  | 1  | 3 |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |    |   |   |    |   |   |   |   |   |   |    |    |    |    |   |    |
| 1    | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2     | 1  | 3  | 2 |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |    |   |   |    |   |   |   |   |   |   |    |    |    |    |   |    |
| 0    | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 3     | 2  | 1  | 0 |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |    |   |   |    |   |   |   |   |   |   |    |    |    |    |   |    |
| 1    | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | -1    |    |    |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |    |   |   |    |   |   |   |   |   |   |    |    |    |    |   |    |
| 1    | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | -1    |    |    |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |    |   |   |    |   |   |   |   |   |   |    |    |    |    |   |    |
| 1    | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | -1    |    |    |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |    |   |   |    |   |   |   |   |   |   |    |    |    |    |   |    |
| 1    | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 1     |    |    |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |    |   |   |    |   |   |   |   |   |   |    |    |    |    |   |    |
| 0    | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 0     |    |    |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |    |   |   |    |   |   |   |   |   |   |    |    |    |    |   |    |
| -1   | -1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | -1    |    |    |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |    |   |   |    |   |   |   |   |   |   |    |    |    |    |   |    |
|      | Do the following:<br>1. Apply convolution operation with kernel A (stride=1, padding = 1) and find the output feature map after applying sigmoid activation function.<br>2. Apply convolution operation with kernel B (stride=1, padding = 1) and find the output feature map after applying sigmoid activation function.<br>3. Apply Max Pool operation on both the outputs obtained from part(1) and part(2) of question 5.<br>4. Analyse the output feature maps obtained after applying convolution operation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |       |    |    |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |    |   |   |    |   |   |   |   |   |   |    |    |    |    |   |    |

CO: Course Outcome

KL: Knowledge Level: L1 – Remember L2 – Understand L3 – Apply L4 Analyze L5 – Evaluate L6 – Create

Course Advisor: Rahul Pal Singh  
 Designation: Assistant Professor